



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

Dataset at a Glance

3,900

Transactions

18

Features

37

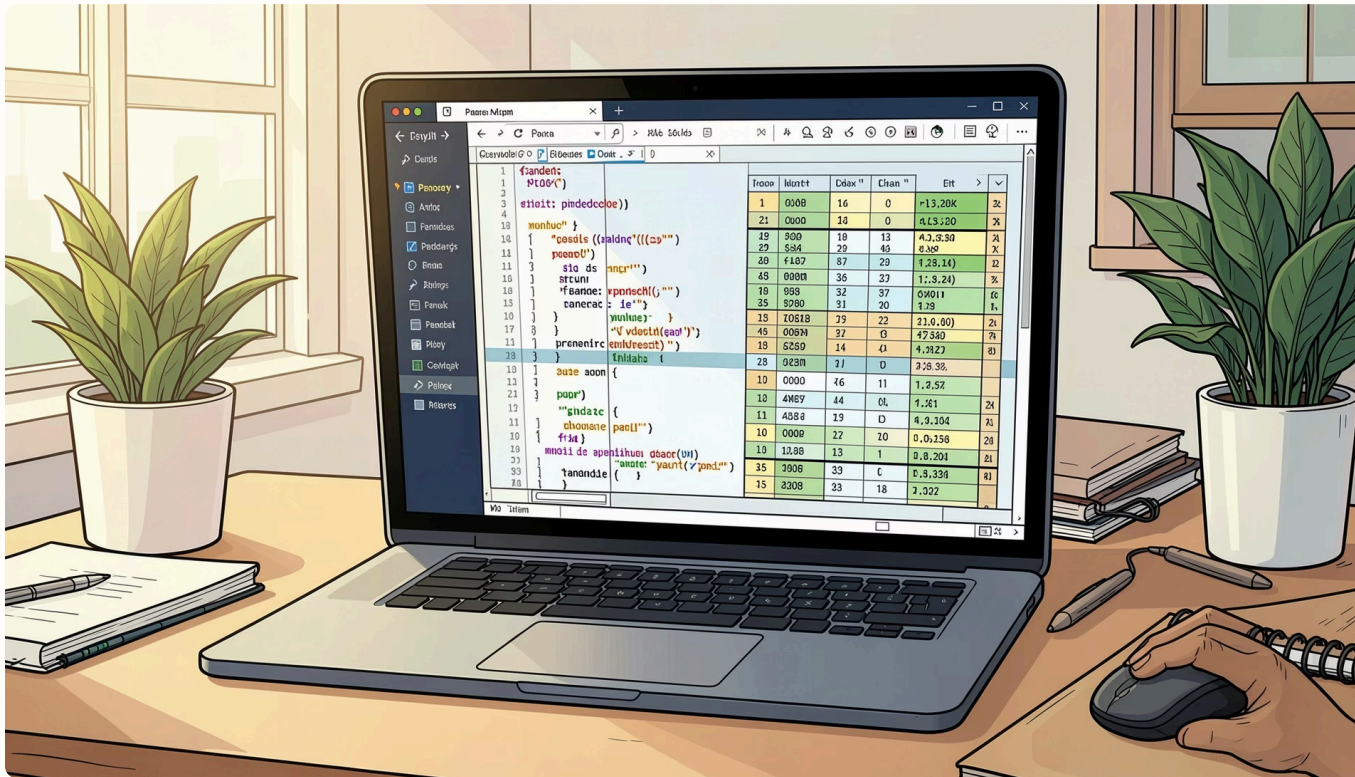
Missing Values

Review Rating column

Demographics · Purchase details · Shopping behavior



Data Preparation in Python



→ Clean & Standardize

Imputed missing Review Ratings by category median; renamed columns to snake_case

→ Feature Engineering

Created `age_group` and `purchase_frequency_days`

→ Database Load

Loaded cleaned DataFrame into SQL SERVER for SQL analysis

Revenue by Gender

Male

\$157,890 total revenue

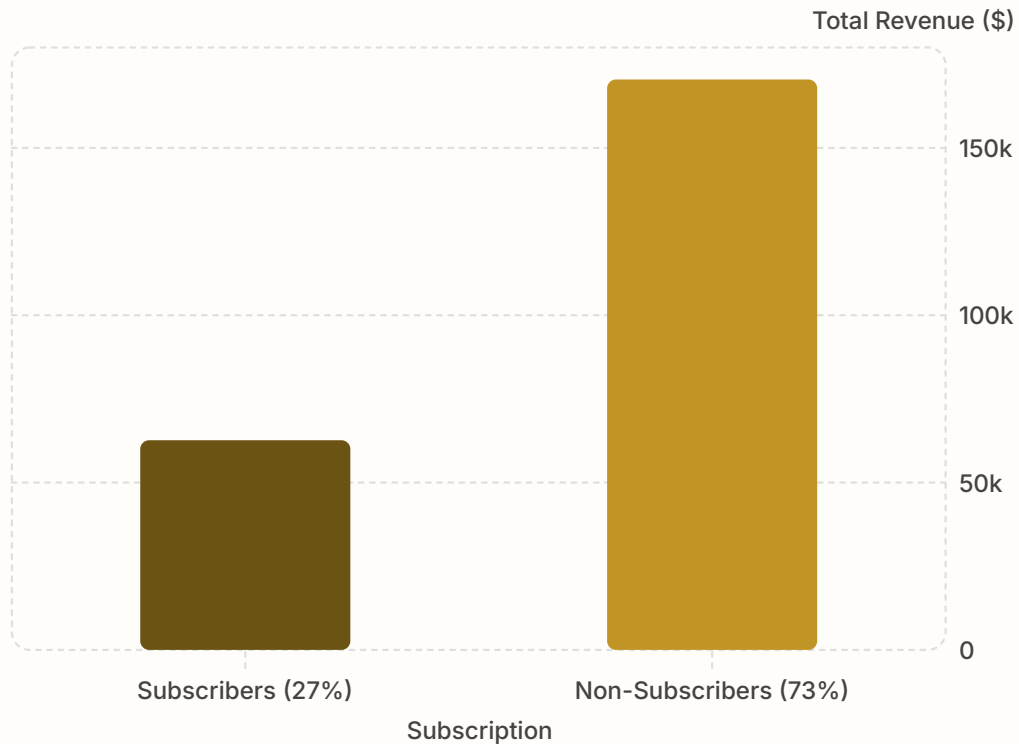
Female

\$75,191 total revenue

Male customers generated **2.1× more revenue** than female customers



Subscribers vs. Non-Subscribers



Key Finding

Non-subscribers drive **73% of customers** and **73% of revenue**

Avg spend is nearly identical: **\$59.49** vs **\$59.87**

i 958 repeat buyers subscribe vs. 2,518 who don't

Customer Segmentation

 Loyal

3,116 customers

 Returning

701 customers

 New

83 customers



Products & Discounts

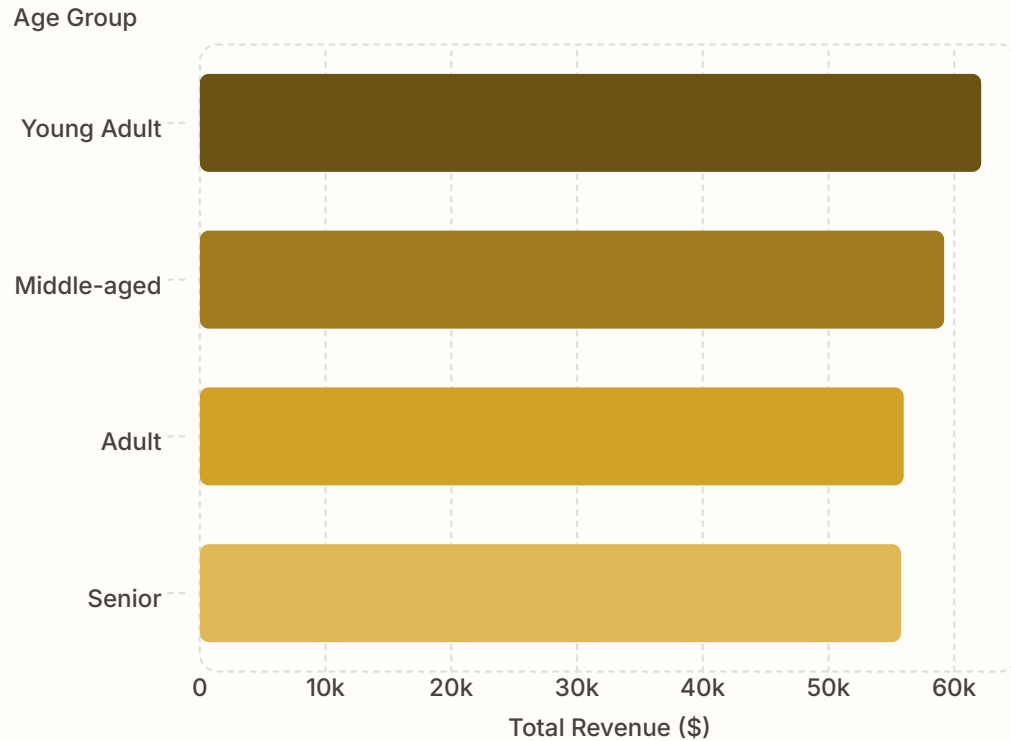
Top 5 by Review Rating

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discounted Products

Item	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

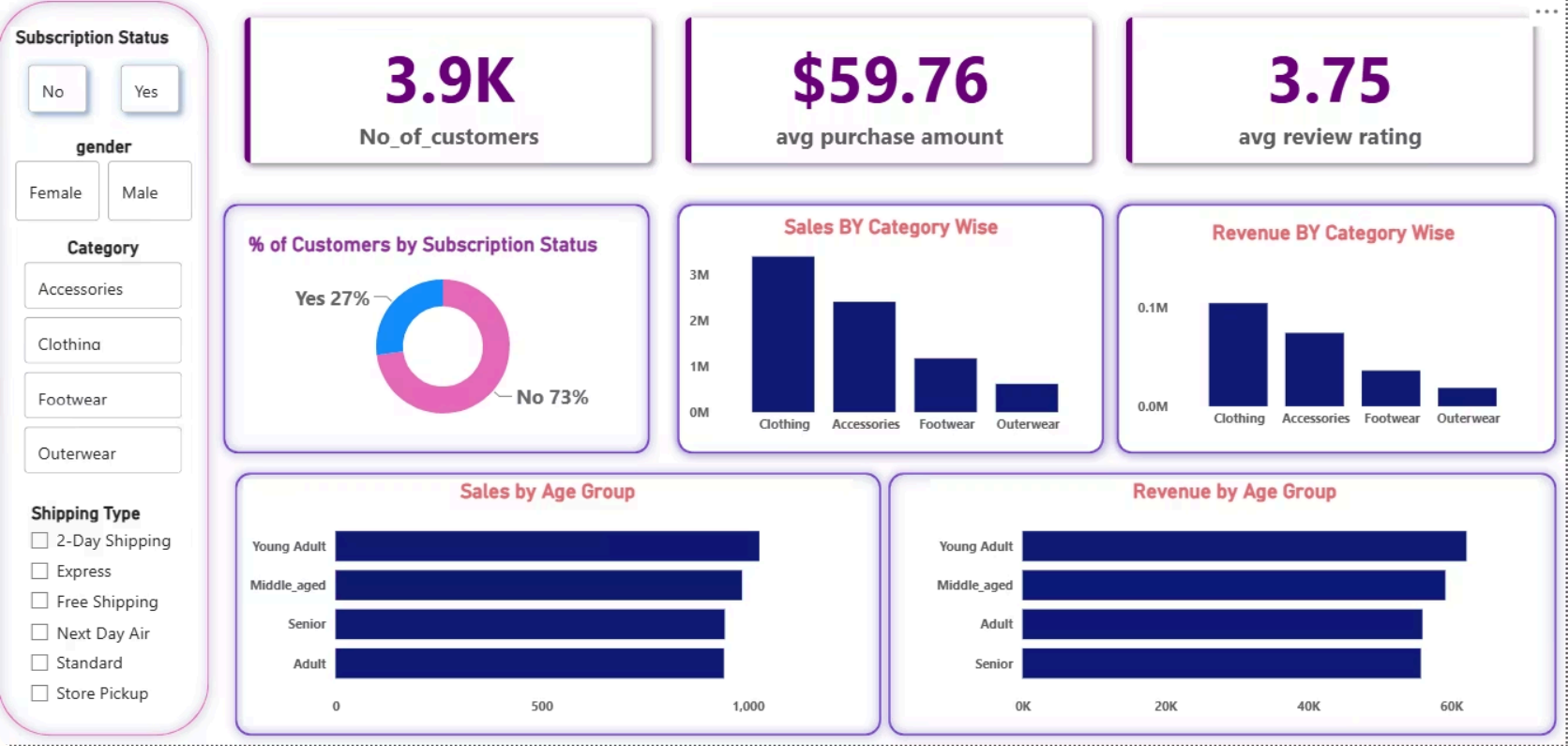
Revenue by Age Group



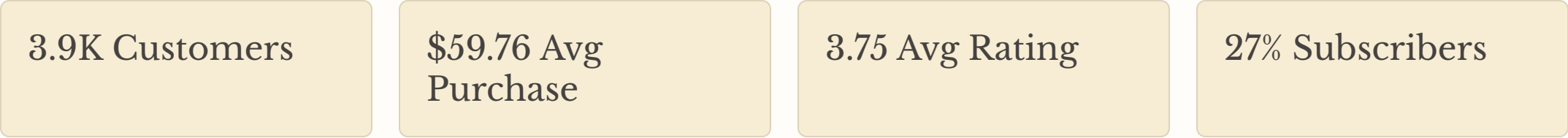
Insight

Young Adults lead revenue at **\$62,143**. Revenue is relatively evenly distributed across all age groups — a **flat \$6K spread** from top to bottom.

Customer Behavior Dashboard



Power BI Dashboard



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

